



Multidisciplinary Spacecraft and Trajectory Uncertainty Quantification for an Interplanetary Mission

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Low tolerance to failure in space systems makes uncertainty quantification vital to its design and optimization. In this work, a sequential multidisciplinary analysis framework is created to optimize and quantify the uncertainties associated with the spacecraft and its trajectory. The engine of the spacecraft is modelled using the rocket propulsion analysis (RPA) tool, and the trajectory is modelled using NASA's general mission analysis tool (GMAT). The selected mission is an interplanetary transfer trajectory from Earth to Mars using B-plane targeting. The analysis and optimization framework OpenMDAO is used to build a multidisciplinary model within which RPA and GMAT are integrated in to sequentially analyze the spacecraft engine and trajectory and then optimize it. The uncertainty associated with the launch date is propagated to the mission trajectory analysis and a Monte Carlo simulation is used to quantify the uncertainty in the spacecraft's time of flight, velocities, and fuel consumed. For a 20-day launch window, the variation in the velocity was computed to be 2-4 km/s and the variation in the fuel consumed was 300 kg. Uncertainty in the engine performance parameters resulted in fuel consumption variations of 0.25 kg during the TCM burn and 30 kg during the MOI burn.

Nomenclature

AR	=	expansion ratio
A_t	=	area of the throat (m^2)
B_R	=	radial B-plane coordinate (km)
B_T	=	tangential B-plane coordinate (km)
C_{thrust}	=	thrust coefficient
F	=	thrust (N)
I_{sp}	=	specific impulse (s)
m_{burn}^i	=	fuel consumed by burn event (kg)
$m_{\text{burn, tot}}$	=	total fuel consumed (kg)
m_{engine}	=	engine mass (kg)
m_{excess}	=	excess fuel (kg)
m_{f_0}	=	initial fuel mass (kg)
P_c	=	chamber pressure (MPa)
MR	=	mixture ratio
R_{mag}	=	final capture orbit radius from center of Mars (km)
TOF	=	time of flight (days)
Δv	=	impulse per unit spacecraft mass (km/s)
v_{sc}	=	spacecraft velocity (km/s)

Subscripts and superscripts:

TCM	=	trajectory correction maneuver
MOI	=	Mars orbit insertion
MF	=	Mars' inertial frame
EF	=	Earth's inertial frame

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I. Introduction

UNDERSTANDING the uncertainties associated with space systems is vital due to the inherent complexity and numerous sources of variability involved. Space missions are exposed to various uncertain parameters that impact the performance, reliability, and lifetime of these systems. These uncertainties can arise from aleatoric factors and epistemic factors, such as launch date uncertainties, manufacturing tolerances, and imprecise models. Uncertainty quantification supports cost-effective decision-making by providing a comprehensive understanding of the trade-offs between performance, reliability, and cost. It allows for the assessment of different design alternatives and the evaluation of their sensitivity to uncertainties, enabling the identification of optimal design choices that balance the performance objectives and cost constraints. Furthermore, the low tolerance of failure for space missions makes it vital to be able assess the likelihood of mission success, predict the performance of spacecraft and instruments, and determine the need for contingencies. Current approaches to multidisciplinary spacecraft design often disregard the associated uncertainties which can lead to unfavorable optimal designs that are vulnerable to the variation in mission conditions. This paper proposes a framework to quantify the uncertainties associated with a multidisciplinary spacecraft and trajectory design by forward propagating the uncertain parameters.

The design and optimization of individual spacecraft systems has been well explored. Fakoor *et al.* [1] conducted a study to optimize the structural system of a space by designing an adaptive optimization framework using genetic algorithms (GAs) with a multi-objective function. The framework conducted finite element analysis to find the optimal structural configuration and payload layout for a remote sensing spacecraft. Similar optimization studies for the spacecraft layout and configuration have been conducted [2, 3]. Galski *et al.* [4] developed a generalized external optimization algorithm (GEO) for the thermal control subsystem. They optimized thermal configuration comprising of five radiators and one battery heater for a Brazilian multi-mission platform. The algorithm designed non-intuitive, new and more efficient designs. A similar study conducted by Muraoka *et al.* [5], also used an GEO algorithm to stochastically optimize the thermal subsystem of the Brazilian multi-mission platform. It incorporated a simpler model to determine the optimum configuration of radiators and solar absorbers with low computational cost.

Okninski *et al.* [6] created an optimization and sizing framework for a bipropellant liquid rocket engine. The optimization algorithm was specifically developed to design satellites using environmentally friendly propellants. During the optimization, the model considers the engine sizing and the thermodynamic properties of the combustion. Several other rocket engine optimization studies have also been conducted over the years [7–10]. A study by Sabug Jr *et al.* [11], the attitude control system was optimized by a black-box optimization (BBO) formulation which simultaneously learns and optimizes the objective function and constraints by using design of experiments. The formulation was applied to a 3U CubeSat with both active and passive attitude control. The BBO formulation designed a system with the best attitude tracking performance. The optimization of a spacecraft's trajectory has been extensively explored in the past few decades. Genetic algorithms in particular are an effective optimization method for maneuver planning and targeting problems [12–15].

Spacecraft are inherently highly multidisciplinary and require multiple subsystems to be coupled for an accurate analysis. A study by Arya *et al.* [16], created a framework that simultaneously co-optimizes the trajectory and the electric propulsion subsystem. The spacecraft is optimized using a composite smoothing control (CSC) framework for a low-thrust maneuver to the asteroid Dionysus.

Uncertainty quantification plays a crucial role in space systems due to the inherent complexity and numerous sources of variability involved. Space missions are exposed to various uncertainties that can impact the performance, reliability, and safety of the systems. These uncertainties can arise from factors such as unpredictable environmental conditions, imprecise models and simulations, manufacturing tolerances, and human errors. Therefore, accurately understanding and quantifying these uncertainties is of utmost importance in ensuring the success of space missions. Shen *et al.* [17], formulated a framework for the robust design of the attitude control subsystem. The formulation used an adaptive control technique and accounted for uncertainties in the attitude tracking of the spacecraft due to actuator faults, imprecise fault estimation, and external disturbances. The simulation results showed that accounting for the uncertainties improved the effectiveness of the overall control strategy. Similar studies on the optimization of spacecraft attitude control systems under uncertainty have been conducted [18–20]. Harris *et al.* [21], designed a coupled spacecraft system and trajectory optimization framework similar to the deterministic optimization formulation of this paper. The study also used the same tools for the subsystem analyses and integrated them into OpenMDAO [22]. They showcase multiple test cases for an Earth-Mars trajectory during the year 2020 with varying optimization configurations. The results indicated that the coupling of subsystems and greater freedom in the design space yielded more optimal designs.

In this paper, a multidisciplinary analysis (MDA) framework is developed for the sequential design analysis and optimization of a spacecraft engine and its trajectory. The MDA model represents the spacecraft system and its trajectory,

leading to a potentially more accurate representation of the uncertainty and design space. This framework has been implemented for an interplanetary mission on an Earth-Mars transfer trajectory using B-plane targeting. OpenMDAO functions as the fundamental infrastructure of the framework, facilitating the integration of the external tools for the engine and trajectory components. Wrappers were developed to seamlessly integrate external tools into the framework, leveraging their accurate modeling capabilities for the individual components. The rocket propulsion analysis (RPA) [23] software tool is incorporated within the engine component, specifically utilized for the design and analysis of the spacecraft's propulsion system. Similarly, the general mission analysis tool (GMAT) [24] software is incorporated within the trajectory component. It utilizes the engine parameters to propagate the spacecraft through the mission sequence and simultaneously verifies adherence to the trajectory constraints. The framework is first formulated as a deterministic optimization problem to determine the optimal design for the multidisciplinary system. Subsequently, this design serves as a basis within the framework to showcase its practical application in uncertainty quantification. It achieves this by systematically propagating the uncertainties associated with the input variables and quantifying the resultant uncertainties observed in the output variables. The uncertainties due to variable launch date and variable engine performance parameters are considered. In conclusion, there exists a significant gap in the exploration of quantifying uncertainty within the design of multidisciplinary space systems. The objective of this paper is to address this problem.

The subsequent sections will delve into further detail regarding the implementation of these tools and the specific problem formulations. The following section describes the test case problem implemented into the framework. The methods section extensively details the MDA framework, optimization formulation, uncertainty quantification formulation and tools used within the MDA framework. In the results section, the optimal design and quantified uncertainties are discussed. The conclusion section summarizes the study's findings and outlines future intended work.

II. Problem Description

In this study, the selected test case involves a spacecraft navigating an Earth-Mars transfer trajectory. The test case is similar to the MAVEN mission [25] and the deterministic optimization formulation conducted by Harris *et al.* [21]. Currently, the onboard systems of the spacecraft solely comprise the engine subsystem; however, additional subsystems will be incorporated in the future. The hypothetical spacecraft incorporates an engine component that is modeled as a chemical rocket engine. Furthermore, the engine is designed as a bipropellant rocket engine using liquid oxygen oxidizer and liquid hydrogen fuel (hydro-lox). The mission sequence of the spacecraft entails an Earth-Mars transfer trajectory with B-plane targeting and two burn events. The burn events are currently modelled as impulse burns for simplification but will be modelled as finite burns in future work to achieve an even more accurate representation of the trajectory.

B-Plane targeting is a fundamental technique used in mission design for planetary flybys or rendezvous. It enables spacecraft to follow a specific hyperbolic trajectory past a planet or enter a capture orbit with a specific inclination. The B-plane represents a reference plane specific to a celestial body, in this instance, Mars. It is a two-dimensional plane perpendicular to the celestial body's orbital plane and passes through its center. This plane is characterized by two orthogonal vectors: the radial vector, extends from the celestial body's center to the location where the spacecraft's trajectory intersects its orbital plane, and the velocity vector, indicates the direction of the spacecraft's velocity relative to the celestial body. The intersection of these vectors defines the B-plane and is often described using coordinate parameters (B_R , B_T). The design constraints for the trajectory can be mapped to these, presenting an advantageous approach for optimization. The desired B-plane coordinates for this problem are $B_R = 7000$ km and $B_T = 0$ km.

These coordinates ensure the spacecraft has a polar capture orbit with an inclination of 90 degrees. The first burn event is a trajectory correction maneuver (TCM) aimed at achieving the desired B-plane coordinates of Mars. Once the TCM is completed, the spacecraft is propagated along the calculated trajectory until it reaches the periapsis of the capture orbit around Mars. This maneuver ensures that the spacecraft has efficiently propagated from Earth to the vicinity of Mars. The second burn event is a Mars orbit insertion (MOI) maneuver, crucial for capturing the spacecraft into Mars' orbit. The desired final orbit for this test case has an altitude of 12,000 km from the center of Mars. Following the second burn event, the spacecraft is propagated until it reaches the apoapsis of the capture orbit. This completes the planned trajectory of the spacecraft from Earth to elliptical orbit around Mars.

The solver iteratively adjusts the delta velocities until it converges on a solution for which the TCM impulse burn attains the desired B-plane coordinates, and the MOI impulse burn achieves a feasible capture orbit. The orbital flight path of the spacecraft is depicted in Fig. 1. The spacecraft's mission sequence entails several key steps. After separating from the launch vehicle, it begins with a 15-day propagation from Earth to reach the TCM location. Following this, the TCM burn is applied. Subsequently, the spacecraft proceeds to propagate until it reaches the periapsis of the capture orbit around Mars. Upon reaching this point, the MOI burn is applied. Finally, the spacecraft continues its journey by

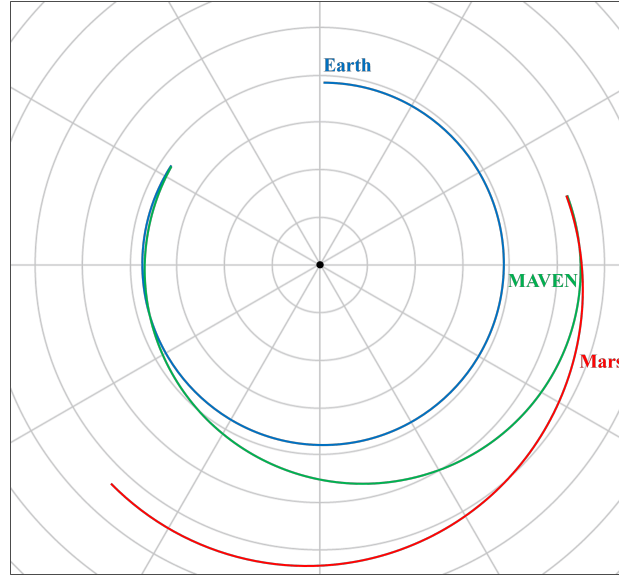


Fig. 1 Spacecraft trajectory from Earth to Mars.

propagating to the apoapsis of the capture orbit around Mars, completing the mission sequence.

III. Methods

The framework for sequential multidisciplinary design analysis and optimization utilizes three tools: OpenMDAO, RPA, and GMAT to model the spacecraft and its trajectory. The problem is initially formulated as a deterministic optimization problem. The optimal design is then utilized as a basis to propagate the uncertainty associated with the input domains. The two uncertain inputs of interest are: the launch date and the engine's performance parameters. These inputs are propagated separately to assess their individual impacts. The subsequent subsections will delve into further detail regarding the implementation of the aforementioned tools and the specific problem formulations involved.

A. Multidisciplinary analysis framework

The multidisciplinary analysis (MDA) is implemented in OpenMDAO [22]. Figure 2 is a graphical representation that depicts how each component is connected to and interacts with other components, which helps in understanding the overall architecture and interdependencies of the system. The OpenMDAO framework for the sequential systems analysis of the spacecraft and trajectory contains two components. The engine component has been specifically developed to model the behavior and characteristics of the spacecraft's chemical rocket engine. The Python wrapper integrates RPA into OpenMDAO, providing a streamlined interface to incorporate the engine's complex thermodynamics calculations and simulations into the overall analysis. The trajectory component solves for a feasible trajectory that satisfies the predefined mission sequence. The Python wrapper developed for the trajectory serves as an interface to execute GMAT through its command-line utility for the trajectory analysis. By leveraging GMAT's capabilities through the wrapper, the model can efficiently compute and optimize trajectories based on the mission's objectives and constraints. By combining the engine and trajectory components within the OpenMDAO framework, an interconnected system for analyzing the spacecraft's onboard systems and trajectory concurrently is established. This integration enables comprehensive MDA, allowing for the evaluation of various factors, such as engine performance, trajectory feasibility, and adherence to mission requirements.

B. Deterministic optimization problem formulation

In this study, the MDA framework is initially utilized to perform deterministic optimization for the coupled system. Within OpenMDAO, an optimization algorithm is used on the MDA framework. The algorithm simultaneously optimizes the spacecraft and trajectory while ensuring the imposed constraints are satisfied. Figure 3 is a graphical representation of the optimization algorithm that depicts how the optimizer interacts with the MDA framework.

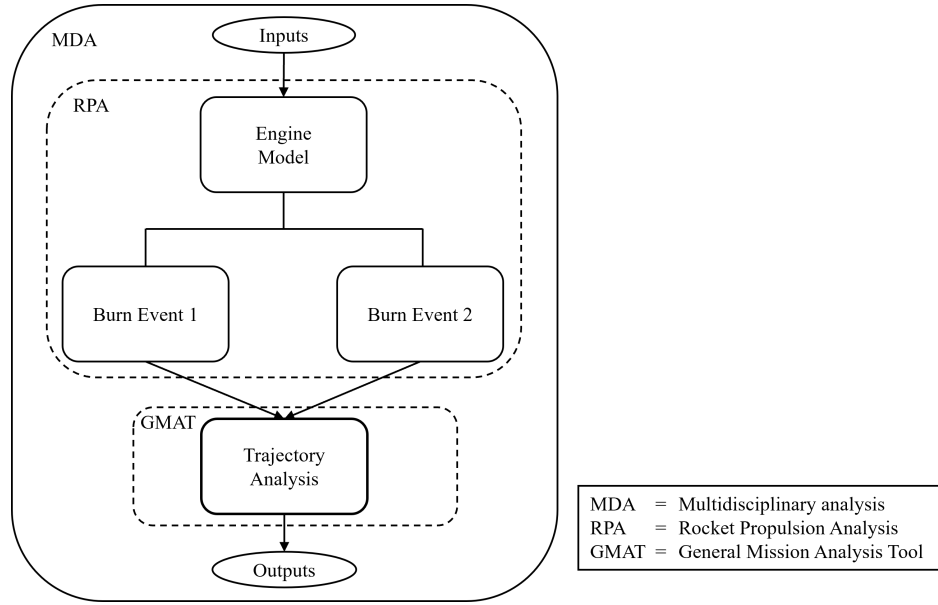


Fig. 2 Multidisciplinary analysis flowchart.

A mathematical representation of the optimization problem formulation is given in Table 1. The formulation consists a total of thirteen design variables. The design variables for the engine component include the chamber pressures (P_c^{TCM} , P_c^{MOI}), mixture ratios (MR^{TCM} , MR^{MOI}), expansion ratio (AR), and area of the throat (A_t) for the engine. Since the two burn events are modeled independently, the engine design variables for chamber pressure and mixture ratio are doubled, while the expansion ratio and area of the throat remain fixed for an engine. The design variables for the trajectory component include the engine performance parameters, the initial fuel mass (m_{f_0}), and the delta velocities of the two burn events (Δv_i^{TCM} , Δv_i^{MOI}).

The objective function of this formulation is defined as the total fuel consumed ($m_{\text{burn, tot}}$) during the two burn events as computed by GMAT. By minimizing fuel consumption, the study aims to find the optimal spacecraft design and trajectory that yield the lowest mission cost. To ensure the feasibility of the optimized sequential design, seven constraint equations are imposed. The first two constraints ensure that the TCM maneuver correctly target the radial and tangential B-plane coordinates of Mars. The third constraint ensures that the final capture orbit of the spacecraft has an altitude of 12,000 km from the center of Mars. The fourth constraint guarantees that the excess fuel remaining after both burn events is at least 200 kg, allowing for sufficient reserve fuel for contingencies, scientific experiments, and potential course corrections. The last three constraints limit the final velocities of the spacecraft in Mars' inertial frame, preventing it from overshooting Mars and ensuring stable capture orbit around Mars. The optimal design obtained through this problem formulation serves as the basis for the subsequent subsection.

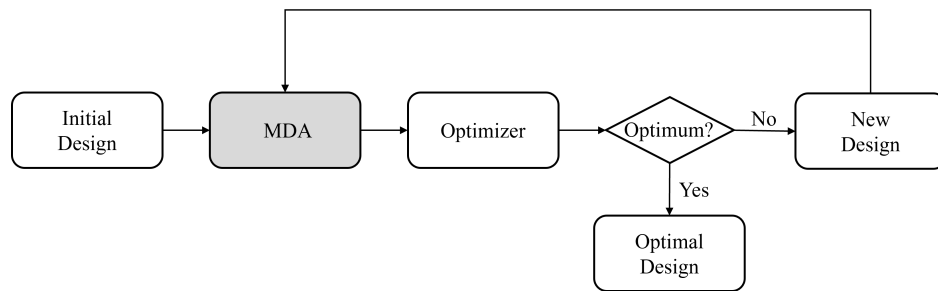


Fig. 3 Deterministic optimization flowchart with multidisciplinary analysis (MDA).

Table 1 Deterministic optimization formulation.

	Variable	Description	Bounds	Quantity
Minimize	$m_{\text{burn, tot}}$	Total fuel burned (kg)		1
w.r.t	P_c^{TCM}	Chamber pressure for TCM (MPa)	[1, 10]	1
	MR^{TCM}	Mixture ratio for TCM	[2, 10]	1
	P_c^{MOI}	Chamber pressure for MOI (MPa)	[1, 10]	1
	MR^{MOI}	Mixture ratio for MOI	[2, 10]	1
	AR	Expansion ratio	[10, 1000]	1
	A_t	Area of throat (m ²)	[10 ⁻⁵ , 10]	1
	m_{i_0}	Initial fuel mass (kg)	[300, 2000]	1
	Δv_i^{TCM}	Δ Velocity for TCM (km/s)	[10 ⁻¹⁰ , 10 ³]	3
	Δv_i^{MOI}	Δ Velocity for MOI (km/s)	[10 ⁻¹⁰ , 10 ³]	3
	Total design variables			13
Subject to	$B_R = 7000$	Radial B-plane coordinate (km)		1
	$B_T = 0$	Tangential B-plane coordinate (km)		1
	$R_{\text{mag}} = 12000$	Final orbit altitude from center of Mars (km)		1
	$m_{\text{excess}} \geq 200$	Excess fuel (kg)		1
	$ v_{\text{sc}}^{\text{MF}} \leq 5$	Spacecraft velocity in Mars's inertial frame (km/s)		3
	Total constraints			7

C. Uncertainty quantification problem formulation

The launch date or departure epoch of a mission is inherently subject to uncertainty due to the unpredictability of various factors. These factors include technical issues, weather conditions, availability of launch vehicles, and the overall readiness of the spacecraft and launch facilities. Technical issues such as unexpected delays in manufacturing, integration, and testing can impact the readiness of the mission for launch. Unfavorable weather conditions at the launch site can also pose constraints and cause launch delays, such as high winds, storms, or hail. Additionally, the availability and scheduling of launch vehicles and launch facilities need to be considered, as they are very limited in availability. It is important when designing space missions to account for this uncertainty to mitigate its impact. Quantifying this uncertainty and propagating it forward provides vital information required for developing contingency plans or incorporating flexibility into the launch schedule. Understanding this uncertainty is also important to determine appropriate launch windows within which desired performance parameters can be designed to be insensitive.

The JPL Horizons ephemeris system is an on-line solar system database and ephemeris computation service [26] that provides solar system data and highly accurate ephemerides for solar system objects. Utilizing the Horizons System, the ephemerides of Mars were computed to determine the distance between Earth and Mars over a three-decade time frame. This distance data and past mission launches have been plotted on Fig. 4. This figure illustrates the periodic variation in the distance between Earth and Mars. The optimal launch window for a Mars mission as defined by minimum-energy requirements, occur every 780 days, corresponding to Mars's synodic period with respect to Earth. Furthermore, the globally minimum-energy launch window varies on an approximately 16-year cycle [27]. A deviation from the optimal launch date of just 1 day can lead to an increased distance of up to approximately 2 million km. Considering the resultant uncertainty in fuel requirements and orbital positions is essential for designing a system that is efficient and reliable. This further emphasizes the importance of understanding the uncertainty associated with the launch date.

To propagate the uncertainty associated with the launch date, a launch window spanning 20 days is considered, with the nominal launch date at its center [28]. The spacecraft's initial conditions are dependent on Earth's ephemerides, which describe the planet's positions and velocities. These ephemerides naturally change as Earth orbits around the Sun, resulting in variations in the spacecraft's initial conditions. By propagating the uncertainty associated with the

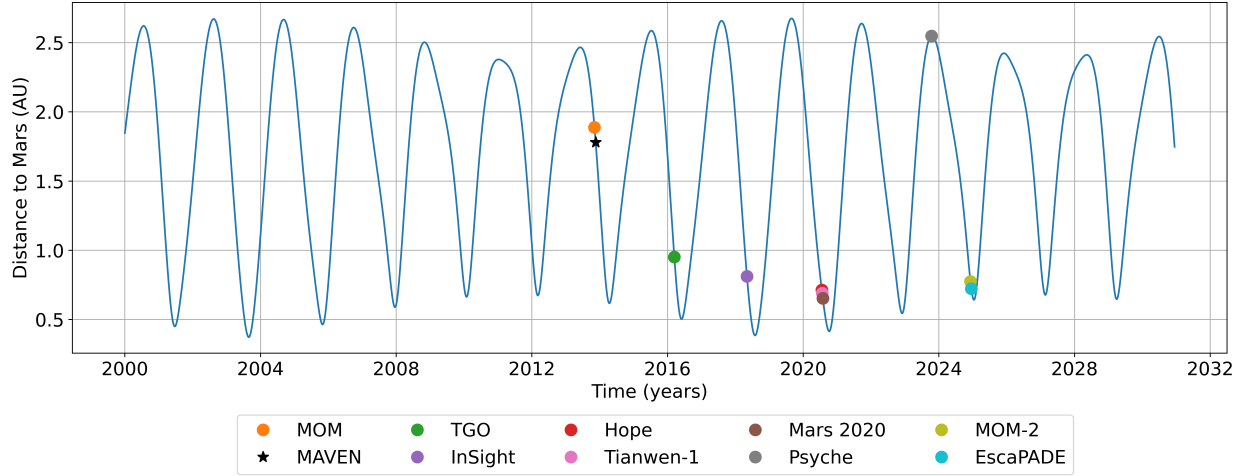


Fig. 4 Launch dates of Mars missions.

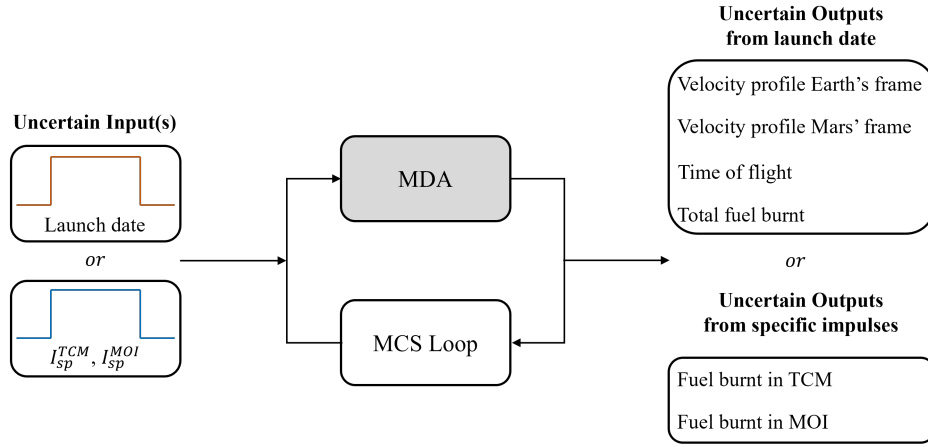


Fig. 5 Uncertainty propagation of launch date using Monte-Carlo simulation (MCS) and multidisciplinary analysis (MDA).

launch date forward, it becomes possible to quantify the uncertainties related to the desired outputs. This forward propagation of uncertainty is accomplished through a Monte-Carlo simulation (MCS) with the MDA framework, as illustrated on Fig. 5. In the algorithm, the uncertainty in the launch date is represented by a random uniform distribution, ranging from 8-28 November, centered on the nominal date of 18 November, 2013. The MCS quantifies the propagated uncertainties in velocity profiles (v_{sc}^{MF}, v_{sc}^{EF}), fuel consumed ($m_{burn, tot}$), and the time of flight (TOF).

The trajectory's design heavily relies on the capabilities inherent in the spacecraft's engines. Conventionally, the evaluation of rocket engine performance occurs within certified facilities specifically designed to accurately test these types of engines. These facilities are equipped with instrumentation calibrated to certified standards and feature highly precise data acquisition systems. Nevertheless, despite the implemented measures, there remains a degree of uncertainty regarding the engine's performance parameters. This uncertainty may stem from diverse sources of error, predominantly arising from the methodologies employed in the testing and measurement processes. Information regarding the uncertainty margins for the performance of an engine analogous to the one used in this specific problem test case is available from a study conducted by Smith *et al.* [29]. The thrust measurement's uncertainty, attributed to instrumentation and calibration, is quantified at $\pm 0.78\%$, while the uncertainty associated with the mass flow rates is quantified at $\pm 0.81\%$. It was established that the thrust uncertainty was the dominating factor, resulting in higher specific impulse (I_{sp}) uncertainty as thrust levels decreased. The root of the sum of squares (RSS) estimate for the I_{sp} uncertainty remained within $\pm 2.5\%$ for the majority of thrust levels. This $\pm 2.5\%$ uncertainty margin is used to define

the uncertainty of the specific impulses for the uncertainty quantification formulation.

In order to incorporate this uncertainty, the I_{sp} for the TCM and MOI events were first stochastically sampled within a defined range of 400 to 450 seconds. This range was established based on the observed I_{sp} values in the deterministic optimization formulation. Subsequently, an uncertainty margin of $\pm 2.5\%$ was applied to the sampled I_{sp} values. This uncertainty is similarly propagated through a MCS with the MDA framework, as illustrated on Fig. 5. For I_{sp} uncertainty, the MCS quantifies the propagated uncertainties in fuel consumed by each burn event ($m_{burn}^{TCM}, m_{burn}^{MOI}$).

D. Rocket propulsion analysis

The rocket propulsion analysis (RPA) tool [23] is a software application designed for the analysis and design of rocket propulsion systems. It serves as a comprehensive tool for simulating and evaluating the performance characteristics of various types of rocket engines. RPA offers a range of capabilities to model and analyze chemical rocket engines. It integrates advanced algorithms and numerical methods to calculate output parameters such as thrust, specific impulse, fuel consumed, nozzle performance, and more. These calculations consider factors such as combustion chemistry, nozzle flow characteristics, and propellant properties.

Within the OpenMDAO framework, the engine component facilitates the calculation of engine performance parameters. It achieves this by passing the relevant design variables to the Python wrapper designed for RPA. The Python wrapper interacts with RPA through its command-line utility, enabling the execution of the engine's thermodynamic analysis. Subsequently, the wrapper returns the analysis results and provides them to the engine component for further processing within the OpenMDAO framework. Integrated into the RPA wrapper is the template input script required by RPA. This script is a CFG (configuration) file and contains the relevant engine configuration data necessary for conducting the RPA analysis. The wrapper formats the template script, to reflect the engine configuration designed by OpenMDAO. By dynamically the CFG file to match OpenMDAO design, the wrapper integrates RPA for a multidisciplinary systems analysis.

The engine component within the system is specifically modeled twice, considering each burn event in the trajectory separately. The engine component formulation can be seen in Table 2. For each instance, the component requires four inputs: chamber pressure (P_c), mixture ratio (MR), expansion ratio (AR), and area of the throat (A_t). To obtain the engine's performance parameters, the first three inputs are sent to RPA for analysis and the wrapper returns the output data for the engine's specific impulse (I_{sp}) and thrust coefficient (c_{thrust}). The other outputs of the component are the thrust (F) and engine mass (m_{engine}). After the RPA analysis, the thrust (F) of each burn event is calculated as

$$F_i = c_{thrust} P_c A_t, \quad (1)$$

where c_{thrust} is the thrust coefficient output, (P_c) is the chamber pressure design input and A_t is the area of the throat design variable. From the thrust (F) of each burn event, its contribution to the portion of engine mass can be calculated as [30]

$$m_i = 1.866 \times 10^{-10} F_i^2 + 0.00130 F_i + 77.4. \quad (2)$$

By adding the mass contributions from both the thrusts (F_1, F_2), the total engine mass is calculated as

$$m_{engine} = m_1 + m_2, \quad (3)$$

where m_1 is the engine mass contribution from the first burn event (TCM) and m_2 is the engine mass contribution from the second burn event (MOI).

E. General mission analysis tool

The general mission analysis tool (GMAT) [24] is an open-source space mission analysis software developed by NASA. GMAT can accurately model the complex dynamics of the spacecraft, plan optimal maneuvers, and design trajectories for a desired mission sequence. Several real missions have extensively utilized GMAT including the Mars Atmosphere and Volatile Evolution (MAVEN) mission by NASA. The MAVEN spacecraft was designed to study the Martian atmosphere and its interactions with the Sun. GMAT was used to simulate the spacecraft's interplanetary trajectory from Earth to Mars. The mission sequence includes two maneuvers or burn events: a trajectory correction maneuver (TCM) and Mars orbit insertion (MOI).

Within the OpenMDAO framework, the trajectory component determines a feasible trajectory for the spacecraft in accordance with the mission sequence and subsequently propagates the spacecraft along this trajectory. To accomplish this, GMAT employs its own script language, in which essential resources, solvers, mission design parameters, and

Table 2 Inputs and outputs of the engine component.

	Variable	Description	Quantity
Inputs	P_c	Chamber pressure (MPa)	1
	MR	Mixture ratio	1
	AR	Expansion ratio	1
	A_t	Area of throat (m ²)	1
	Total inputs		4
Outputs	I_{sp}	Specific impulse (s)	1
	F	Thrust (N)	1
	m_{engine}	Mass of engine (kg)	1
	Total outputs		3

Table 3 Inputs and outputs of the trajectory component.

	Variable	Description	Quantity
Inputs	m_{f_0}	Initial fuel mass (kg)	1
	m_{engine}	Mass of engine (kg)	1
	I_{sp}^{TCM}	Specific impulse for TCM (s)	1
	Δv_i^{TCM}	Δ Velocity for TCM (km/s)	3
	I_{sp}^{MOI}	Specific impulse for MOI (s)	1
	Δv_i^{MOI}	Δ Velocity for MOI (km/s)	3
	Total inputs		10
Outputs	TOF	Time of flight (days)	1
	m_{burn}^i	Fuel consumed by each burn event (kg)	2
	m_{excess}	Excess fuel (kg)	1
	v_{sc}^{EF}	Spacecraft velocity profile in Earth's inertial frame (km/s)	3
	v_{sc}^{MF}	Spacecraft velocity profile in Mars' inertial frame (km/s)	3
	B_R	Radial B-plane coordinate (km)	1
	B_T	Tangential B-plane coordinate (km)	1
	R_{mag}	Final orbit altitude from center of Mars (km)	1
	Total outputs		13

desired outputs are defined. The engine performance parameters computed by the engine component are then transmitted to the trajectory component as inputs for the sequential design process. To facilitate this integration, a wrapper is developed that incorporates a modified GMAT script and can format it with the engine performance parameters and trajectory inputs. Subsequently, the wrapper executes the modified script in GMAT using its command-line utility. By dynamically updating the GMAT script with the relevant inputs, GMAT is effectively integrated into this OpenMDAO component. This seamless integration between the components enables multidisciplinary design analysis and optimization of the spacecraft's onboard systems and trajectory. The trajectory component formulation can be seen in Table 3.

The modified GMAT script incorporated into wrapper defines the necessary resources and establishes the desired mission sequence for the analysis. The script begins by defining the spacecraft and its initial conditions including a hardware component for the fuel tank which is needed to compute the fuel consumption. Subsequently, the script defines the two burn events, with their specific impulses that are computed from the engine component. To solve for the trajectory, a differential solver resource and its corresponding conditions are defined. To simulate the different stages of the trajectory, the script establishes propagators for the three specific sections: near Earth, deep space, and near Mars. The desired outputs must also be defined within the script, these include velocity profile plots and orbital flight path plots derived from multiple reference points. Additionally, the wrapper provides time series data concerning objectives, constraint functions, and other variables for analysis. Once all resources are created in the script, the mission sequence is carefully defined to ensure the desired trajectory is achieved, meeting the mission's objectives and constraints. GMAT exports the aforementioned data to a text file at each iteration of the mission. The wrapper subsequently analyzes this file and transmits the data to the OpenMDAO framework. The wrapper can dynamically change the engine parameters and other trajectory inputs, but the mission sequence remains fixed. To use the wrapper for a different trajectory test case, a new mission sequence must be written and incorporated into the wrapper.

IV. Results

In this section, the results of the problem formulations within the multidisciplinary framework are presented. First, the results of the deterministic optimization problem are discussed. Subsequently, the results derived from the forward propagation of uncertainties within the framework are examined. This involves an analysis of the trends and distributions characterizing the uncertain output parameters. Specifically, the study focuses on the uncertainties associated with the launch date and the specific impulses of the two burn events.

A. Deterministic design optimization

The sequential multidisciplinary spacecraft and trajectory system is deterministically optimized at a fixed nominal launch date of November 18th, 2013. The resulting optimal design for the system is presented in Table 4. The starting point design presented in the table does not satisfy any constraints and consequently, fails to achieve a satisfactory final capture orbit around Mars. Whereas the optimal design performs exceptionally better and successfully achieves a satisfactory final capture orbit. The optimal design achieves a minimized fuel consumption of 502 kg and completes the mission with a total time of flight of 307 days. The specific impulse of the engine converges around 440 s for both burn events. The analysis of the Δv_z^{MOI} values suggests that in order to minimize fuel consumption, trajectories with burns concentrated along one or two axes are preferred over a distributed burn across all three. During the trajectory's two burn events, a minimal quantity of fuel is expended during the TCM event. Whereas, the predominant portion of the fuel is utilized during the MOI event, primarily aimed at decelerating the spacecraft sufficiently to enable its entry into a capture orbit around Mars.

Figure 6 illustrates the velocity profiles of the optimized spacecraft from Earth's inertial frame and Mars' inertial frame. From Figure 6b, it is observable that the spacecraft's velocity components in Mars' inertial frame decrease substantially as it approaches Mars. This substantiates the confirmation that the spacecraft has successfully entered into a capture orbit around Mars. This optimized design and velocity profile serve as the basis for the subsequent uncertainty quantification analyses.

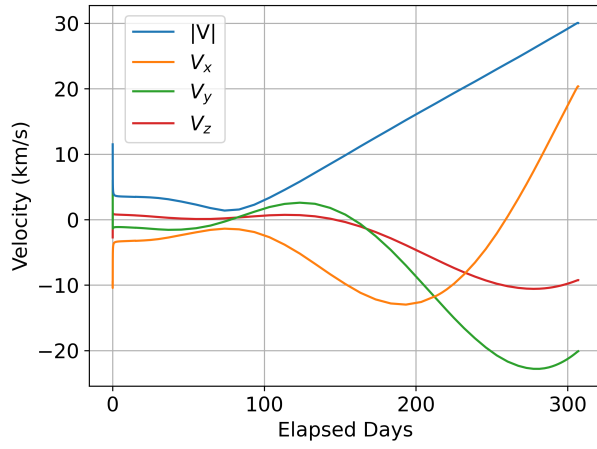
B. Quantifying uncertainty due to variable launch date

The MCS algorithm was performed using 1000 iterations of the randomly sampled launch date to quantify the impact of its uncertainty on the output variables. The propagated uncertainty in velocity profile of the spacecraft was quantified at each time step of the mission sequence. These velocity profiles from Earth's inertial frame and Mars' inertial frame, for the velocity components (x, y, z) and the magnitude, are illustrated in Fig. 7. These velocity margins also provide other insights into the consequences of launch date uncertainties. Notably, for a launch window of 20 days, the velocity components in the Earth inertial frame exhibit deviations ranging from 2-4 km/s at any given instance. The margin for the magnitude of the velocity remains relatively consistent, while the y and z components demonstrate stronger fluctuations.

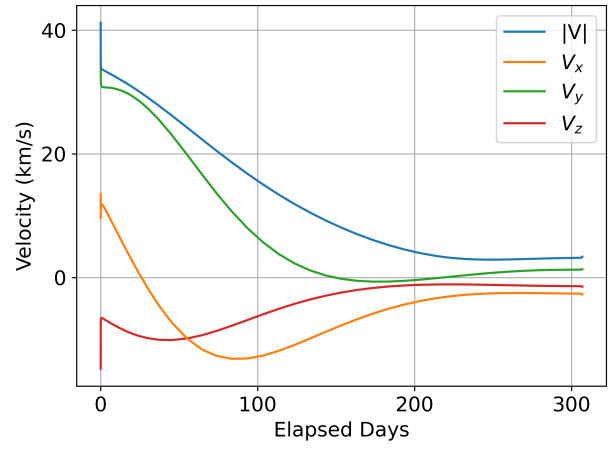
It is notable that the velocity margin of the z component from Earth's inertial frame converges around 180 days into the mission. This convergence point coincides with the initiation of the MOI burn event, indicating that it occurs at the same number of elapsed days into the mission for all the feasible trajectories. Indicating that for this mission

Table 4 Deterministic optimization results.

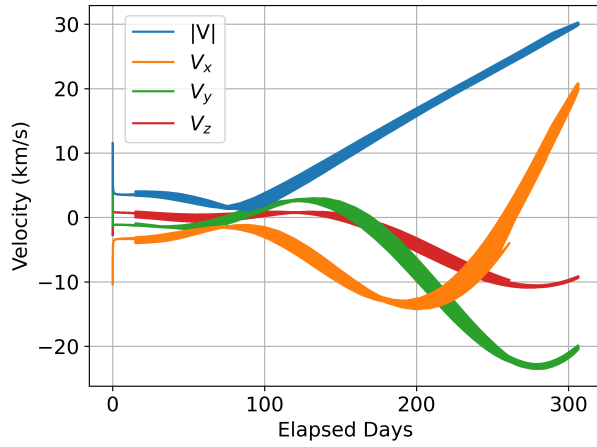
Variable	Starting Point	Optimized Result	Variable	Starting Point	Optimized Result
P_c^{TCM} (MPa)	1	10	P_c^{MOI} (MPa)	1	5.76
MR^{TCM}	1	4.85	MR^{MOI}	1	4.12
AR	1	56.9	A_t	0.001	0.0053
$I_{\text{sp}}^{\text{TCM}}$ (s)	269.57	439.39	$I_{\text{sp}}^{\text{MOI}}$ (s)	269.57	438.80
Δv_x^{TCM} (km/s)	1	0.003938	Δv_x^{MOI} (km/s)	-1	-1.603439
Δv_y^{TCM} (km/s)	1	0.006044	Δv_y^{MOI} (km/s)	0	-0.000001
Δv_z^{TCM} (km/s)	-1	-0.000675	Δv_z^{MOI} (km/s)	0	0.000001
m_{f_0} (kg)	2000	702.25	$m_{\text{burn, tot}}$ (kg)	1835.69	502.25
m_b^{TCM} (kg)	1369.54	2.71	m_b^{MOI} (kg)	466.15	499.54
TOF (days)	824.22	307.271			



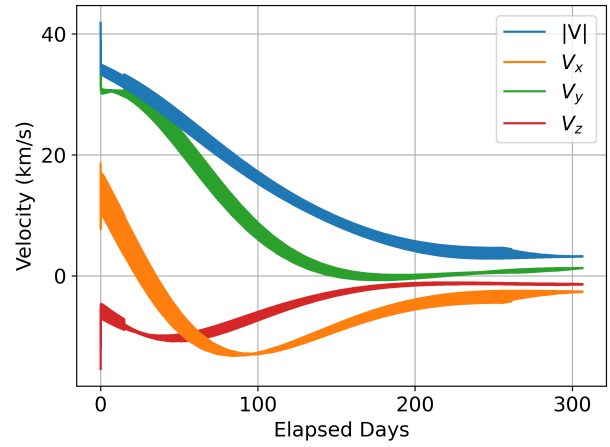
(a)



(b)

Fig. 6 Velocity profile of the deterministic optimal design: (a) Earth's inertial frame, (b) Mars' inertial frame.

(a)



(b)

Fig. 7 Velocity profile with uncertainty: (a) Earth's inertial frame, (b) Mars' inertial frame.

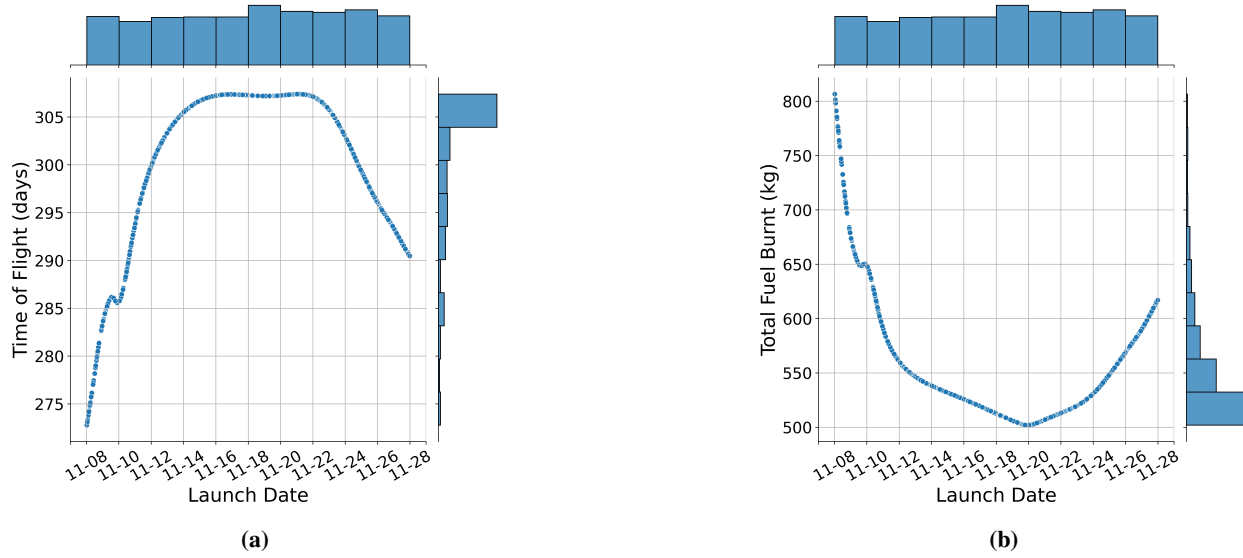


Fig. 8 Marginal histograms with the launch date input; (a) time of flight, (b) fuel consumption.

sequence, the instance when the MOI burn occurs depends on the spacecraft's position and exhibits insensitivity to both its velocity and launch date. From Fig. 7b, it is observable that the final velocity components in Mars' inertial frame maintain a magnitude below 5 km/s, indicating that the imposed constraint is appropriately satisfied even when the input uncertainty is propagated forward.

The propagation uncertainty of the launch date also yields notable trends in the time of flight (TOF) and total fuel consumed ($m_{\text{burn, tot}}$), as depicted in Fig. 8. The marginal histograms reveal a continuous trend for the time of flight and total fuel consumption. For a 20 day launch window, the variation in the time of flight was computed to be 30 days and the variation in the total fuel consumed was 300kg. Both parameters exhibit a unimodal frequency distribution with opposing skews, suggesting they are inversely related. The depicted scatter plots demonstrate that the time of flight parameter exhibits a bell-shaped probability distribution and has a maximum near the nominal launch date. Conversely, the total fuel consumed exhibits a bimodal probability distribution with a minimum at the nominal launch date. Deviation from the nominal launch requires significantly higher fuel, which is expected as the distance between Earth and Mars is at its minimum at this date. The time of flight is at its peak near nominal launch date but begins to decrease significantly beyond an 8-day window. Suggesting that due to the increased distance to Mars, the spacecraft must travel faster to be able to effectively target Mars and satisfy the trajectory constraints. As a consequence, the spacecraft completes the mission sooner, but at the cost of a substantial increase in fuel consumption. Additionally, as the deviation from the nominal launch date increases, there is an exponential increase in the computational cost required by the framework to identify a feasible trajectory. This increased computational cost is dependant on the initial guess used by the framework.

C. Quantifying uncertainty due to variable engine performance parameters

The MCS algorithm was similarly performed using 1000 iterations of the randomly sampled engine performance parameters ($I_{\text{sp}}^{\text{TCM}}$, $I_{\text{sp}}^{\text{MOI}}$) within a defined range of 400 to 450 seconds and a $\pm 2.5\%$ uncertainty margin. The velocity profiles of the spacecraft and its time of flight exhibited a notable insensitivity to these uncertainties for this formulation.

The propagated uncertainty of burn events with respect to their engine performance parameters are depicted in Fig. 9. The depicted scatter plots demonstrate the linear relation between the I_{sp} and the fuel consumed. The computed variation in the fuel consumed during the TCM burn amounted to 0.25 kg or $\pm 5\%$, whereas variation in the MOI burn amounted to 30 kg or $\pm 3\%$. This computed variation in the fuel consumed remained consistent across the entire I_{sp} range. When analyzing the fuel consumption holistically, the uncertainty associated with the TCM burn is insignificant as the majority of fuel is consumed during the MOI burn.

The quantified fuel margins are important consideration to take into account in the design of the spacecraft's fuel tank and in formulating the reserve fuel constraints. The quantified fuel margins are important consideration to take into

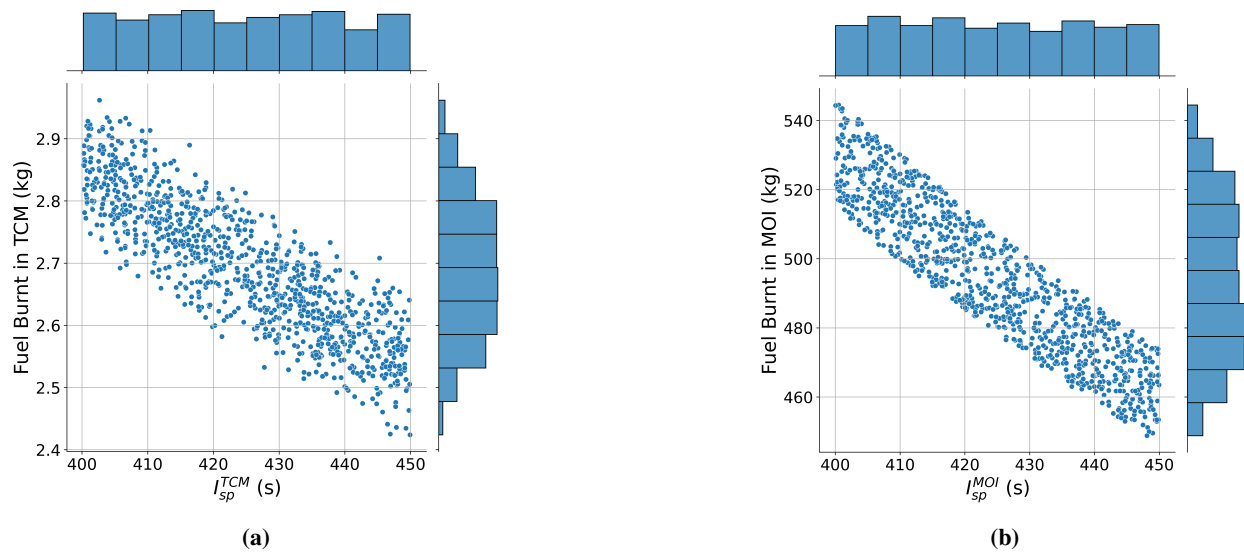


Fig. 9 Marginal histograms with the launch date input; (a) time of flight, (b) fuel consumption.

account when designing the fuel tank of the spacecraft and formulating the reserve fuel constraints.

V. Conclusion

In this paper, a multidisciplinary design analysis and optimization framework was developed for the spacecraft's onboard system and trajectory. This model was deterministically optimized, and the optimal case was used as the basis for the uncertainty quantification. The launch date and engine performance parameters were introduced as uncertain parameters and, by forward propagation, the system's uncertainty was quantified. For a 20-day launch window, the variation in the velocity profiles was computed to be 2-4 km/s and the variation in the fuel consumed was 300 kg. The probability distribution of the fuel consumption and time of flight was quantified with respect to the launch date. By propagating the uncertain engine performance parameters the computed variation in the fuel consumed amounted to 0.25 kg for the TCM burn and 30 kg for the MOI burn. Future work includes implementing the framework for design under uncertainty, such as reliability-based design or robust design optimization. Additionally, to improve the computational feasibility, surrogate models of this system will be built into the framework.

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